ECHO Case Report

The Quantathon, Qubits Qrew, NED University Muhammad Usman Murtaza · Stretch module chosen: 02 (The Honest Coin) and 04 (The Erased Measurement)

1. Verdict

ECHO was not a faster-than-light message. Dr. Mahmud's detectors **agreed**, but agreement is correlation, not communication. I am about **90% confident** of this physics conclusion, because it follows from standard quantum theory and my own runs (Section 3). I am much less confident about *who* was involved. Nothing I recovered identifies a person, so I treat the human question as open.

The most likely story is that Zara mistook a correlation for a signal, and that her "random" trigger seed was biased (about 70/30), so ECHO did not fire by pure chance.

2. Evidence by module

Module	What I did	Result	What it proves
01 Gatekeeper	X on q0, H on q1; H then H; X on q0 and q2	q0 always 1, q1 about 50/50; H,H always 0; vault pattern 101 in 100/100 shots	I can build certain, uncertain and "fake-uncertain" bits. H,H is certain because the paths to 1 cancel.
02 Honest Coin	H coin; the recovered seed circuit `ry(1.159)`	Coin: 517 vs 483. Seed: 719 zeros vs 305 ones (about 70/30)	The seed is **not fair**. Bias is about 0.20 (P(1) near 0.30 instead of 0.50).
03 Two Observers	Bell pair (H, CNOT); pre-agreed secret; H on both before measuring	Bell: only 00 and 11. Secret: also only 00 and 11.	"Always agreeing" does not prove a quantum link. Section 3 gives the test that does.

Module	What I did	Result	What it proves
04 Erased Measurement	H,H; H,measure,H; X,H,H; H,X,H	H,H always 0; with a mid-circuit measurement about 50/50; X,H,H always 1; H,X,H always 0	Gates are reversible, measurement is not. Interference does the "erasing".
05 Damaged Message	Majority vote over 32 triples	Decoded to the ASCII word **ECHO**	Repetition corrects 1 flip per triple, detects 2, and fails on 2 (110 read as 1 when the truth was 0).

Bias number (Module 02). A fair coin over n shots has standard error $0.5/\sqrt{n}$. For a bias of 0.20, being 3 standard errors away needs $n \approx (3 / (2 \times 0.20))^2 \approx$ **56 shots**. My 1024-shot run was about 12 standard errors away, so this is not an unlucky fair coin.

3. The crux: agreed, or could they send a message?

They only agreed. There are two separate arguments.

(a) Agreement alone proves nothing. Two people who flip a coin beforehand and each read out the result also never disagree. In my Module 03 runs that fake gave 00 and 11 with the same histogram as the Bell pair.

(b) No message can travel anyway (no-signalling). Bob's qubit is 50/50 no matter what Alice does to hers. I checked this on a Bell pair. Bob's P(0) was 0.500 when Alice did nothing, 0.501 when she measured in Z, and 0.524 when she measured in X (4000 shots each, so the last difference is within normal noise). Since Bob's data look the same under every choice Alice makes, Alice cannot encode anything for him to read. Teleportation (Module 06 circuit) also needs two classical bits, so it never beats light speed.

What does separate quantum from a pre-agreed secret? A weak test is to measure both qubits in the H basis. The real Bell pair still gives only 00 and 11, while a secret with a fixed value per qubit gives about 25% for each outcome (my rerun of 300 rounds gave 83, 75, 64, 78 for 00, 01, 10, 11). But a cleverer classical cheat could pre-agree both an answer for the Z basis and an answer for the H basis, so this test is **not conclusive**. The conclusive test is CHSH, which uses different measurement angles on each side. Any pre-agreed

secret scores at most **2**. My Bell pair scored **S = 2.83** (8000 shots per setting, angles 0, π/2 for Alice and ±π/4 for Bob), which is the quantum maximum 2√2. So the link is real, but it still carries no message.

4. Competing theory

The strongest alternative is that ECHO really worked and someone suppressed it, as Prof. Vance or Sana Rao might have wanted. I reject it because the physics forbids the claimed effect: entanglement gives matching random results, and a sender gains no control over them. A second alternative, that the biased seed was a coincidence, is ruled out by the 56-shot calculation above. Who chose the angle 1.159 stays unexplained.

On the "changed when observed" log entry (Module 04), ARIA's excuse is physically possible, because measuring a quantum state really does change it. So that claim alone shows neither tampering nor innocence. It does imply that someone or something present performed a measurement.

5. Limitations and next experiment

- My evidence supports the physics verdict but **does not identify a suspect**. Suspect statements are Claims, not Facts.
- Everything ran on a noiseless simulator, so real hardware could show lower CHSH scores.
- The seed result shows the circuit is biased, not who set the angle.

Next experiment: run the full teleportation circuit (Module 06) with and without the two classical bits, to show the receiver's accuracy drops to about 50% without them, and rerun CHSH on real hardware noise models.

6. Evidence board (summary)

- **Fact:** seed is about 70/30; Bell pair agrees; secret also agrees; CHSH S = 2.83; Bob's marginal stays at 50%; Module 05 decodes to ECHO.
- **Claim:** Zara: entanglement carried a message. ARIA: the log changed because it was observed. Sana: only a corrupted log was deleted.
- **Inference:** ECHO could not signal; the seed was biased, possibly deliberately.
- **Open question:** who set the seed angle; the machine's name and run time from Module 01.

7. References and AI use

AI use: I used Claude (Anthropic) to explain concepts, check my code and draft this report. I ran all circuits myself in Qiskit with the Aer simulator and I can explain and modify each one.

References: Qiskit and Qiskit Aer documentation; Nielsen & Chuang, *Quantum Computation and Quantum Information* (2000); Clauser, Horne, Shimony & Holt, Phys. Rev. Lett. 23, 880 (1969) (CHSH); Wootters & Zurek, Nature 299, 802 (1982) (no-cloning); Bennett et al., Phys. Rev. Lett. 70, 1895 (1993) (teleportation).